

TensorFlow Basics for ML

tensorflow basics for ml using python

If you're learning Machine Learning, **TensorFlow** is one of the most popular Python libraries for building and training deep learning models.

TensorFlow Basics for Machine Learning (Python)

1. What is TensorFlow?

TensorFlow is an open-source library developed by Google for:

- Machine Learning (ML)
- Deep Learning (DL)
- Neural Networks
- Computer Vision
- Natural Language Processing (NLP)
- Time Series Forecasting

2. Installation

Bash
`pip install tensorflow`

Check the version:

Python
`import tensorflow as tf`
`print(tf.__version__)`

3. Import TensorFlow

Python
`import tensorflow as tf`

4. What is a Tensor?

A tensor is the basic data structure in TensorFlow.

Python	TensorFlow
int	Scalar Tensor

Python	TensorFlow
list	Vector Tensor
nested list	Matrix Tensor

Example:

```
Python
x = tf.constant(5)
print(x)
```

Output

```
tf.Tensor(5, shape=(), dtype=int32)
```

Vector

```
Python
x = tf.constant([1,2,3,4])
print(x)
```

Output

```
tf.Tensor([1 2 3 4], shape=(4,), dtype=int32)
```

Matrix

```
Python
matrix = tf.constant([
    [1,2],
    [3,4]
])
print(matrix)
```

5. Tensor Types

```
Python
tf.int32
tf.float32
tf.float64
tf.string
tf.bool
```

Example

```
Python
x = tf.constant(3.14, dtype=tf.float32)
print(x)
```

6. Tensor Shape

```
Python
x = tf.constant([[1,2],[3,4]])
print(x.shape)
```

Output

```
(2,2)
```

7. Creating Tensors

Zeros

```
Python
tf.zeros((3,3))
```

Output

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]
```

Ones

```
Python
tf.ones((2,2))
```

Random Numbers

```
Python
tf.random.normal((3,3))
```

Uniform random

```
Python
tf.random.uniform((3,3))
```

8. Tensor Operations

Addition

```
Python
a = tf.constant([1,2,3])
b = tf.constant([4,5,6])
print(a+b)
```

Multiplication

```
Python
```

```
print(a*b)
```

Matrix Multiplication

```
Python
a = tf.constant([[1,2],[3,4]])
b = tf.constant([[5,6],[7,8]])
print(tf.matmul(a,b))
```

9. Convert Tensor to NumPy

```
Python
x = tf.constant([10,20,30])
print(x.numpy())
```

Output

```
[10 20 30]
```

10. Variables

Unlike constants, variables can change during training.

```
Python
w = tf.Variable(5)
w.assign(10)
print(w)
```

11. Mathematical Functions

Square

```
Python
tf.square([2,3,4])
```

Square Root

```
Python
tf.sqrt([4.0,9.0,16.0])
```

Mean

```
Python
tf.reduce_mean([10,20,30])
```

Sum

```
Python
tf.reduce_sum([10,20,30])
```

Maximum

```
Python
tf.reduce_max([10,20,30])
```

12. Reshaping

```
Python
x = tf.constant([1,2,3,4,5,6])
y = tf.reshape(x,(2,3))
print(y)
```

Output

```
[[1 2 3]
 [4 5 6]]
```

13. Neural Network Module (Keras)

TensorFlow includes **Keras**, a high-level API for building neural networks.

```
Python
from tensorflow import keras
```

Create a simple model:

```
Python
model = keras.Sequential([
    keras.layers.Dense(10, activation='relu'),
    keras.layers.Dense(1)
])
```

14. Compile Model

```
Python
model.compile(
    optimizer='adam',
    loss='mse',
    metrics=['mae']
)
```

15. Train Model

```
Python
model.fit(X_train,
          y_train,
```

```
epochs=10,
batch_size=32)
```

16. Prediction

```
Python
prediction = model.predict(X_test)

print(prediction)
```

17. Save Model

```
Python
model.save("model.keras")
```

Load model

```
Python
model = keras.models.load_model("model.keras")
```

18. Complete Example (Linear Regression)

```
Python
import tensorflow as tf
from tensorflow import keras
import numpy as np

# Data
X = np.array([[1],[2],[3],[4],[5]], dtype=float)
y = np.array([2,4,6,8,10], dtype=float)

# Model
model = keras.Sequential([
    keras.layers.Dense(1, input_shape=[1])
])

# Compile
model.compile(
    optimizer='adam',
    loss='mse'
)

# Train
model.fit(X, y, epochs=500, verbose=0)

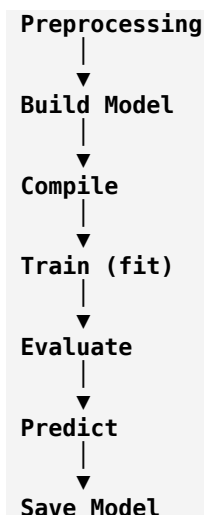
# Predict
print(model.predict([[6.0]]))
```

Expected output (approximately):

```
[[12]]
```

19. TensorFlow Workflow

Dataset
↓
▼



20. Topics to Learn Next

1. Tensor operations and broadcasting
2. Automatic differentiation (**GradientTape**)
3. Keras Sequential API
4. Functional API
5. Activation functions (ReLU, Sigmoid, Softmax)
6. Loss functions
7. Optimizers (Adam, SGD, RMSprop)
8. Regularization (Dropout, L1/L2)
9. Callbacks (EarlyStopping, ModelCheckpoint)
10. Convolutional Neural Networks (CNNs)
11. Recurrent Neural Networks (RNNs) and LSTMs
12. Transfer learning
13. TensorBoard for visualization
14. Model deployment (TensorFlow Lite, TensorFlow Serving)

Practice Project Ideas

- House price prediction (Regression)
- Customer churn prediction (Binary classification)
- Handwritten digit recognition (MNIST)
- Spam email detection (Text classification)
- Image classification (Cats vs. Dogs)
- Stock price forecasting with LSTM

Since you've been learning Python, scikit-learn, and working toward a data science career, TensorFlow is a natural next step after mastering core ML concepts. A recommended progression is:

Python → NumPy → Pandas → Matplotlib → scikit-learn → TensorFlow/Keras → Deep Learning projects → MLOps basics.

ChatGPT can make mistakes. Check important info.